

COM3502/4502/6502 SPEECH PROCESSING

Lecture 14

Articulatory Phonetics

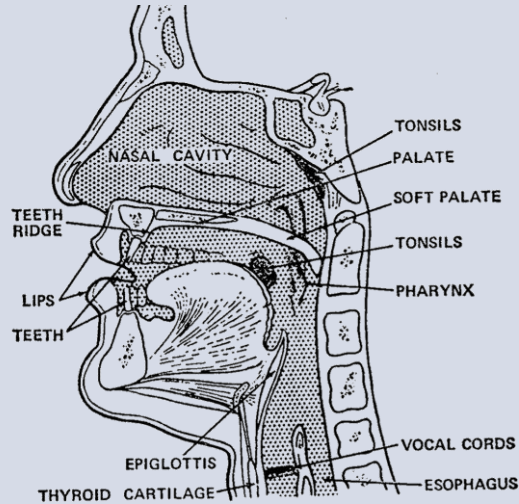


Articulatory Phonetics

... a description of speech sounds in terms of the *physical actions* performed in their production

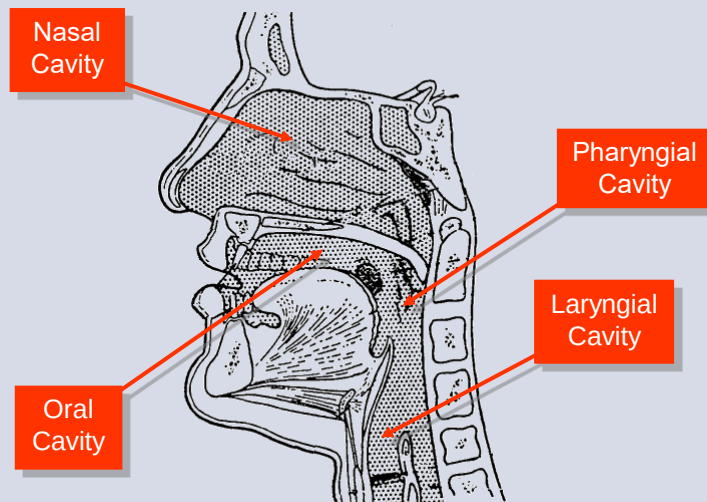


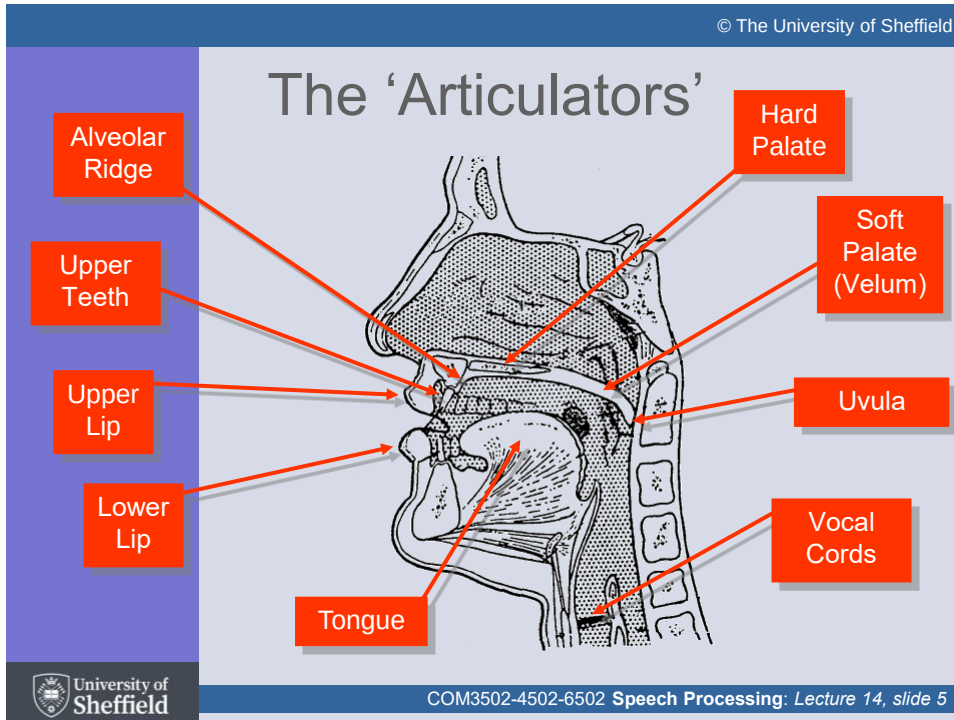
The Vocal Tract – A Reminder



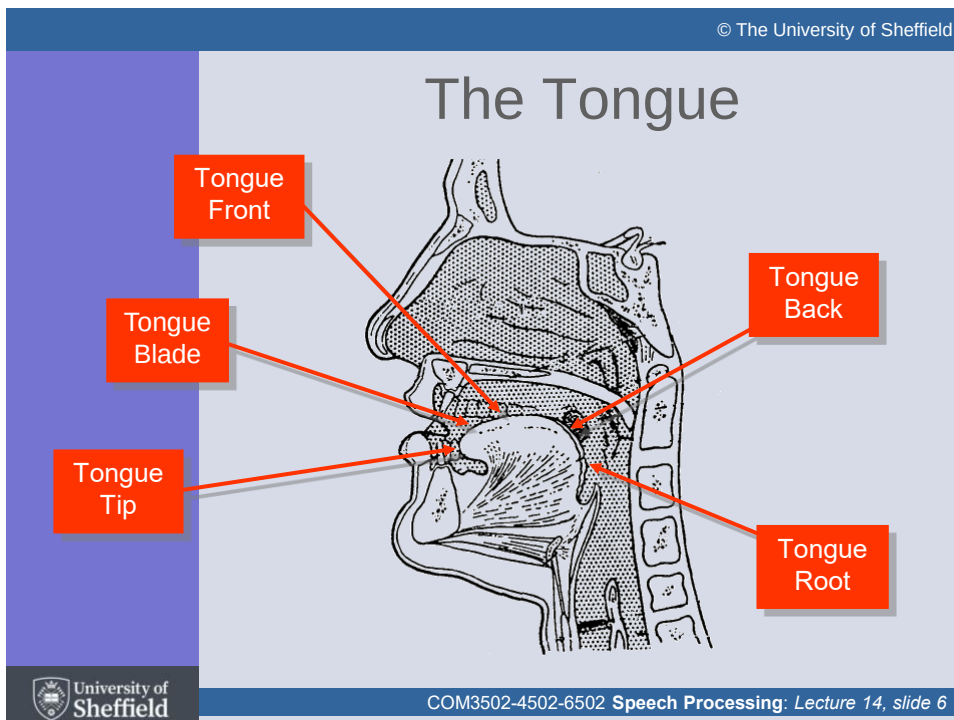
Taken from: Denes, P. B., & Pinson, E. N. (1973). *The Speech Chain: The Physics and Biology of Spoken Language*. New York: Anchor Press.

The Resonant Cavities





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Articulatory Phonetics



Speech sounds are classified in
'**articulatory phonetics**' as follows ...

- vowels & consonants (*i.e. all sounds*)
 - where the air stream comes from
 - whether air is going in or out
- consonants
 - whether the vocal cords are vibrating: '**voice**'
 - where the constriction is: *the* '**place**' of articulation
 - how the sound is made: *the* '**manner**' of articulation
- vowels
 - the position of the tongue
 - the shape of the lips

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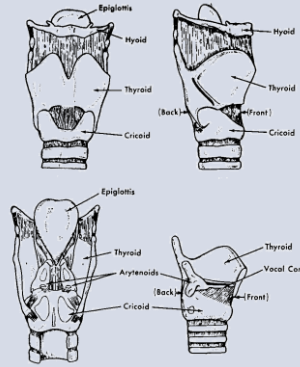
Consonants

Voice - Place - Manner

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Voice

- Covered in Lecture 10 (Speaking)
- ‘Degree of voicing’
 - voiced
 - voiceless
- ‘Voice quality’
 - modal (*normal*)
 - creaky
 - falsetto
 - breathy



Human Larynx

Consonants

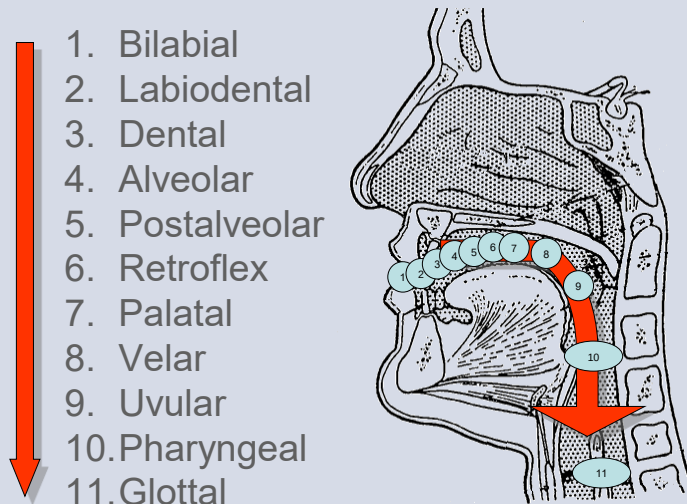
Voice - Place - Manner

'Place' of Articulation



- Articulation refers to the **constriction** of the vocal tract during speech production
- Articulation involves the movement of an '**active articulator**' (e.g. *the tongue*) towards a '**passive articulator**' (e.g. *the top of the mouth*)
- The '**place**' of articulation refers to the physical location of the constriction in the vocal tract

'Place' of Articulation



'Place' of Articulation

Place of Articulation	Active Articulator	Passive Articulator	Example Sounds
Bilabial	upper and lower lips	<i>none</i>	[p b m]
Labiodental	lower lip	upper front teeth	[f v]
Dental	tongue tip	upper front teeth	[θ ð]
Alveolar	tongue tip or blade	alveolar ridge	[t d n l s z]
Postalveolar	tongue tip or blade	rear of alveolar ridge	[ɹ ʃ]
Retroflex	tongue tip	hard palate	[ɭ ɻ]
Palatal	tongue front	hard palate	[j ɲ]
Velar	tongue back	soft palate	[k g ŋ]
Uvular	tongue back	uvula	[q ɢ]
Pharyngeal	tongue root	rear wall of pharynx	[ħ]
Glottal	vocal folds	<i>none</i>	[h ʔ]

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Consonants

Voice - Place - Manner

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'Manner' of Articulation

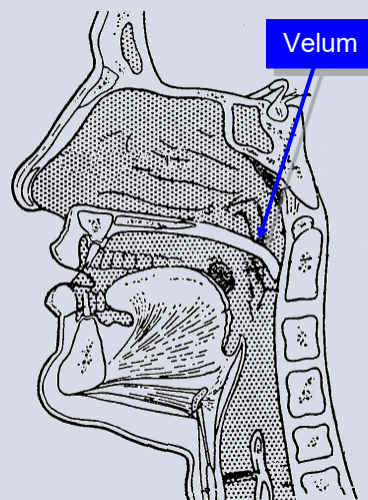


- The '**manner**' of articulation refers to the way in which the airstream is modified by the primary and secondary articulators
- Degrees of '**stricture**' ...
 - closure: articulators in firm contact ('**stops**')
 - narrowing: articulators close together but not touching ('**fricatives**')
 - approximation: wide gap between articulators ('**approximants**')

'Manner' of Articulation

Stops:

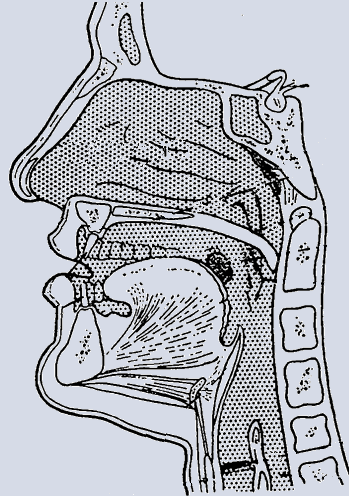
- complete blockage of the airstream
- can be produced at many different places of articulation
- stops made with a velic closure are called '**oral stops**'
- stops made without a velic closure (*and with airflow through the nasal cavity*) are called '**nasal stops**'



'Manner' of Articulation

Oral Stops:

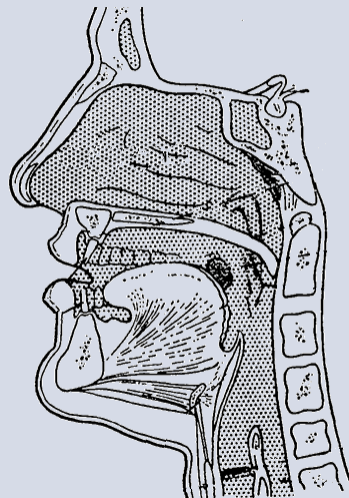
- air pressure rises rapidly in the vocal tract during closure
- the compressed air bursts out of the vocal tract with a small explosive noise when the closure is released
- such sounds are called '**plosives**', e.g. [p t k b d g]
- slower release sounds are called '**affricates**', e.g. [tʃ dʒ]



'Manner' of Articulation

Fricatives:

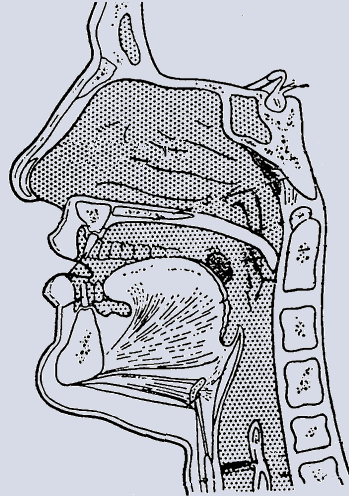
- sound is produced with a very narrow opening between the active and passive articulators
- the resulting airflow is **turbulent** and hence noisy
- air for a '**median fricative**' flows through a narrow channel in the centre of the cavity, e.g. [s]
- airflow for a '**lateral fricative**' escapes through a narrow space at the sides of the tongue, e.g. [ɬ]



'Manner' of Articulation

Approximants:

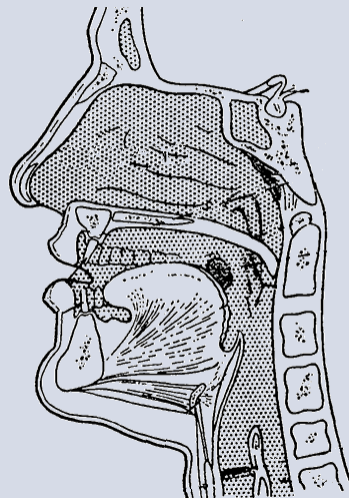
- sonorants do not have a stricture narrow enough to cause turbulence
- 'lateral approximants' have complete closure on the midline, e.g. [l]
- 'median approximants' have no complete closure and are also called 'semi-vowels' or 'glides', e.g. [w ʋ j]



'Manner' of Articulation

Taps and trills:

- closure in a 'tap' or 'flap' is much shorter than for a plosive hence there is no build up of air pressure, e.g. [ɾ]
- a 'trill' or 'roll' consists of a series of rapid tap-like closures, e.g. [r]



Voice, Place, Manner

I
P
A

Manner

Place

CONSONANTS (PULMONIC)											
	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap				ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Where symbols appear in pairs, the one to the right represents a voiced consonant. Shaded areas denote articulations judged impossible.

Voice



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Voice, Place, Manner



- Voice, Place, Manner (**VPM**) labels are the standard method of specifying consonants

- For example ...

- [p] → voiceless bilabial plosive
- [g] → voiced velar plosive
- [s] → voiceless alveolar fricative
- [tʃ] → voiceless postalveolar affricate
- [n] → voiced alveolar nasal
- [t] → voiceless alveolar plosive
- [l] → voiced alveolar lateral-approximant

CONSONANTS (PULMONIC)											
	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap				ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			



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Voice, Place, Manner



	Plosive	Frictive	Nasal	Glottal	Labial	Dental	Alveolar	Palatal	Velar	Glottal
Plosive	p b	t d	k g							
Frictive		f v	s z	ʃ ʒ	θ ð					
Nasal			m	n	ɱ					
Glottal										ʔ

Some more examples ...

- [k] → voiceless velar plosive
- [θ] → voiceless dental fricative
- [m] → voiced bilabial nasal
- [b] → voiced bilabial plosive
- [h] → voiceless glottal fricative
- [ʃ] → voiceless postalveolar fricative
- [ʔ] → glottal stop

The English Consonants

Consonant Pairs

Single Consonants

Unvoiced

- p pea
- t tea
- k key
- f fin
- θ thin
- s sip
- ʃ ship
- tʃ chin
- m map
- ŋ hang
- l led
- w wet

Voiced

- b bay
- d day
- g gay
- v vat
- ð that
- z zoo
- ʒ azure
- dʒ jam
- n nap
- h hit
- r red
- j yet

The (English) Phonetic Chart

Strict IPA for:

- [tʃ] is tʃ
- [dʒ] is dʒ

OTHER SYMBOLS

ɱ Voiceless labial-velar fricative	ɕ ʑ Alveolo-palatal fricatives	Open
ɱ Voiced labial-velar approximant	ɻ Voiced alveolar lateral fricative	
ɥ Voiced labial-palatal approximant	ɧ Simultaneous ʃ and x	
ħ Voiceless epiglottal fricative	Affricates and double articulations can be represented by two symbols joined by a tie bar if necessary.	ts kp
ʕ Voiced epiglottal fricative		
ʁ Epiglottal plosive		

CONSONANTS (PULMONIC)															
	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal				
Plosive	p b			t d			ʈ ɖ	c ɟ	k ɡ	q ɢ	ʔ				
Nasal	m	ɱ		n			ɳ	ɲ	ɴ						
Trill	ʙ			r						ʀ					
Tap or Flap				ɾ											
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ʕ				
Lateral fricative				ɬ ɮ											
Approximant				ɹ			ɻ	j	ɰ						
Lateral approximant				l			ɭ	ʎ	ʟ						

Where symbols appear in pairs, the one to the right represents a voiced consonant. Shaded areas denote articulations judged impossible.

English Consonants

Consonant Pairs

Single Consonants

Consonants

[p]	[b]	[t]	[d]	[k]	[g]	[tʃ]	[dʒ]
[f]	[v]	[θ]	[ð]	[s]	[z]	[ʃ]	[ʒ]
[m]	[n]	[ŋ]	[h]	[l]	[ɹ]	[w]	[j]



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The Glottal Stop ?

CONSONANTS (PULMONIC)															
	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal				
Plosive	p b			t d			ʈ ɖ	c ɟ	k ɡ	q ɢ	ʔ				
Nasal	m	ɱ		n			ɳ	ɲ	ɴ						
Trill	ʙ			r						ʀ					
Tap or Flap				ɾ											
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ʕ				
Lateral fricative				ɬ ɮ											
Approximant				ɹ			ɻ	j	ɰ						
Lateral approximant				l			ɭ	ʎ	ʟ						

Where symbols appear in pairs, the one to the right represents a voiced consonant. Shaded areas denote articulations judged impossible.

English Consonants

Close of the vocal cords (stop breath and sound, then release) as in:

[ʌʔoə] → “uh oh”

Can replace e.g /t/ or /h/ (or other sounds)

[wə:tə] → “Water”

[wə:ʔə] → “Water”



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Vowels

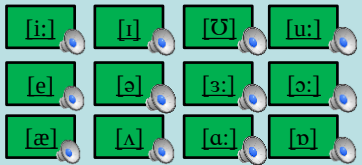


- Vowels are articulated by ...
 - raising the front or the back of the tongue towards the roof of the oral cavity
 - shaping the lips
 - ... thereby changing the 'vowel quality'
- Vowel quality is governed by ...
 - 'vowel height': *high / low*
 - 'vowel location': *front / back*
 - 'lip position': *rounded / unrounded*

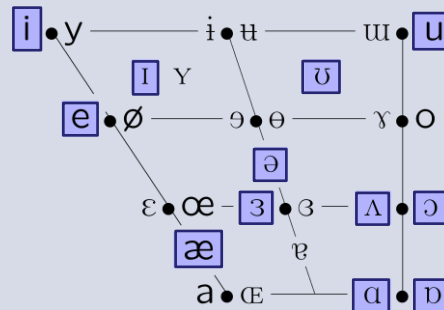


Engl. Vowel Sounds / Monothongs

English Vowels (Monothongs)

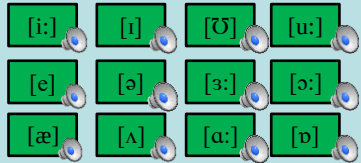


English Vowels

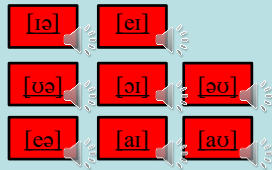


Engl. Vowel Sounds / Diphthongs

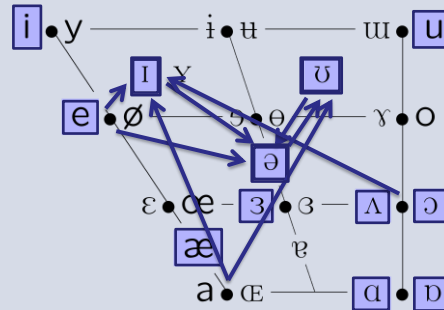
Monothongs (single vowel sound)



Diphthongs (gliding vowels)



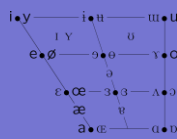
English Vowels



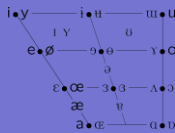
Vowel Height



- The **'height'** of a vowel refers to the relationship between the highest point of the tongue and the roof of the oral cavity
- A **'close'** or **'high'** vowel (such as [i] or [u]) is produced when the tongue is raised close to the roof
- An **'open'** or **'low'** vowel (such as [æ] or [a]) is produced when there is a wide gap between the highest point of the tongue and the roof of the oral cavity
- Vowels can also be **'mid'** ([ə]), **'half-close'** ([e]) or **'half-open'** ([ʌ])



Vowel Location



- The '**location**' of a vowel refers to the part of the tongue which is highest
- A '**front vowel**' (such as [i]) is produced by raising the front part of the tongue towards the hard palate
- A '**back vowel**' (such as [u]) is produced by raising the back part of the tongue towards the soft palate
- A '**central vowel**' is produced by raising the central part of the tongue towards the junction of the hard and soft palates
- The mid-central vowel [ə] is called "**schwa**"

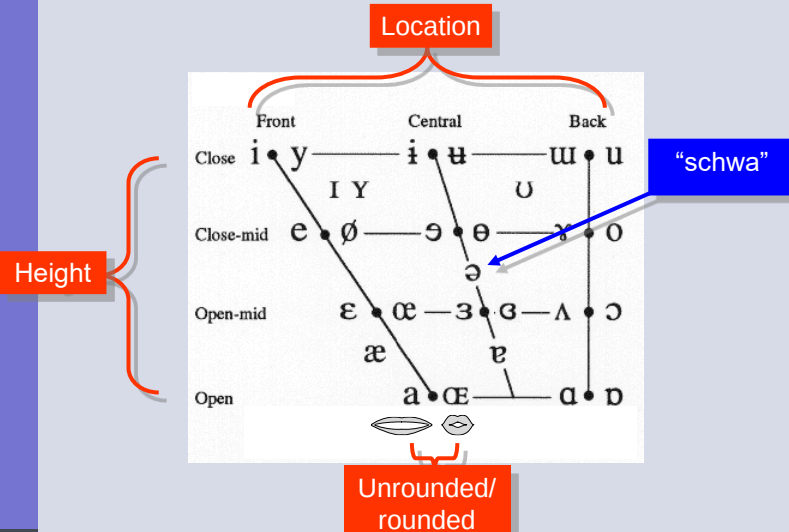


Picture source

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Vowels

I
P
A

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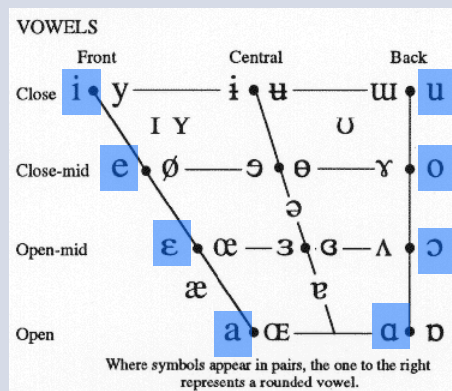
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Vowels



- Many languages make a distinction between ...
 - rounded and unrounded vowels
 - long and short vowels
 - oral and nasalised vowels
 - monophthongs and diphthongs (*vowel glides*)
- Vowel quality can be indicated by ...
 - placing a dot on the ‘**vowel quadrilateral**’
 - ... or ...
 - relating it to a set of language-independent ‘**cardinal vowels**’

The Vowel Quadrilateral



The Cardinal Vowels

Vowel Instability



- Vowels appear to carry less information than consonants, e.g. ...
[ən əpəl ə də kəps ðə dəktə əwə]
- Vowels seem to act as a **carrier signal** that is **modulated** by the consonants
- For these reasons, vowel quality is very variable and can drift over time (*hence giving rise to different historical and contemporary accents*) ...
 - Chaucer (*1340s): “house” → [hu:s]
 - Shakespeare(*1560s): “house” → [hous]
 - Modern day: “house” → [haus]
- This is why a non-phonetic writing system can have an advantage; people can communicate despite having very different accents

This slide set has covered ...

- The articulators
- Consonants
 - voice, place, manner (VPM)
- Vowels
 - vowel quality
 - vowel height (*open-close*)
 - vowel location (*front-back*)
 - rounded/unrounded
 - the neutral vowel ‘schwa’
 - the vowel quadrilateral
 - cardinal vowels

Any Questions ?



Next time ...

Acoustic Phonetics